

COI 362/632 - DBMS - Assignment 1: SQL Queries

- You must use PostgreSQL 15.x for this assignment.
- There are a total of 35 queries in this assignment.

Download the cleaned-up data from this link. It is a zip file that can be imported directly to PostgreSQL. Refer to lecture notes on importing a zip file into PostgreSQL.

Dataset

This dataset is structured into two primary modules—hospital ("hosp") and intensive care unit ("icu")—which together span tens of thousands of patient encounters and aggregate millions of individual records. We highly suggest you also examine the schema and data before writing your queries.

Hospital Module ("hosp")

- Scope:** Derived from a hospital-wide electronic health record, this module covers a broad spectrum of patient care data primarily recorded during hospital stays, with some information extending to outpatient settings.
- Data included:**
 - Patient and admission details (e.g., basic demographics, admissions, transfers)
 - Laboratory measurements and related metadata
 - Microbiology culture results
 - Provider orders and medication administration details
 - Medication prescriptions and pharmacy information
 - Billing data, including diagnostic and procedure codes, billing events, and service-related information
 - Additional service-related records

ICU Module ("icu")

- Scope:** Sourced from a critical information system specific to intensive care, this module focuses on high-resolution ICU data. It has been organized into a flat schema linking ICU stays to a variety of detailed event records.
- Data included:**
 - Patient ICU stays and associated metadata
 - Records of intravenous fluids and medications, including their ingredients
 - Patient outputs, procedures performed, and charted observations
 - Events captured as specific datatime entries
 - Structure: Each event record is linked by stay and item identifiers for detailed traceability across tables.

Description

hosp Module

Table Name	Description
admissions	Information about each hospital admission (hadm_id), including admission and discharge times, admission type, location, etc.
d_hcpcs	Dimension table describing codes used in Hcpcsvents (mostly CPT codes).
d_icd_diagnoses	Dimension table describing ICD-10 diagnosis codes used in diagnoses_icd.
d_icd_procedures	Dimension table describing ICD-10 procedure codes used in procedures_icd.
d_labitems	Dimension table describing laboratory test items. Links to d_labtests.
diagnoses_icd	ICD-10 diagnosis billed for each hospital admission. Links to d_icd_diagnoses.
drugs	Diagnosis-related group (DRG) codes associated with hospital stays.
emar	Electronic Medication Administration Record (eMAR) data, documenting medication administrations (barcodes scanning at bedside).
emar_detail	Detailed breakdown of each eMAR event, including dose given and route.
labevents	Hospital billing events containing HCPCS codes. Links to d_hcpcs.
laborders	Laboratory measurements and results from hospital-wide labs. Links to d_labtests.
microbiologyevents	Microbiology cultures and antibiotic sensitivities.
omr	On-site Medical Record data, often containing outpatient-type measurements (height, weight, etc.).
patients	Core patient-level information (gender, anchor age, date of death).
pharmacy	Detailed history of filled medications, including dosing information and status. Links to prescriptions and emar via pharmacy_id.
pos	Provider Order Entry (POE) records. General interface for entering patient orders (lab, medication, consults, etc.).
pos_detail	Supplemental detail (in an Entity-Attribute-Value format) for entries in pos.
prescriptions	Medications ordered for a patient, including route, frequency, and identifying codes (DSN, NDC). Links to pharmacy and emar.
procedures_icd	ICD-10 procedure codes billed during hospital stays. Links to d_icd_procedures.
provider	Key de-identified provider identifiers used across hospital (e.g., order_provider_id).
transfers	Track clinical services transferred to the patient (e.g., "Hypertension", "Cardiology").
transfers	Tracks physical movement of patients across different wards/units in the hospital.

ICU Module

Table Name	Description
caregiver	De-identified caregiver identifiers for the ICU module.
chartersents	The integrity of ICU charted data (vital signs, ventilator settings, etc.). Links to d_items for definition of items.
datetimeevents	Datetime-type measurements charted in the ICU (e.g., heart rate).
d_items	Dimension table defining terms for ICU measurements.
icustays	Defines individual ICU stays, with admission and discharge times.
ingredientevents	Ingredients used within the same solution for a drug (e.g., multiple additives in an IV bag).
inputevents	Continuous infusions or intermittent administrations (IV fluids, drips, etc.).
outputevents	Output data (urine output, drain volumes) recorded in the ICU.
procedures_icd	Procedures documented during the ICU stay (e.g., mechanical ventilation, imaging orders), also includes scheduled times and caregiver info.

Detailed Tables of Columns

Below is a reference table for each dataset table. Columns appear in the following format:

Column Name	Data Type	Description
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hosp Module

• admissions

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing the patients table (each patient has a unique subject_id).
hadm_id	bigint	Primary key for each hospital admission (unique per admission).
admission_type	text	Date and time the patient was admitted.
dischtime	text	Date and time the patient was discharged.
deathtime	text	If applicable, date and time of in-hospital death (often matches dischtime).
admission_type	text	General classification of the admission's urgency or type (e.g., ELECTIVE, URGENT, etc.).
order_provider_id	text	De-identified ID of the provider who admitted the patient (links to provider_id in the provider table if needed).
admission_location	text	Location of the patient before arriving (e.g., emergency room).
discharge_location	text	Location to which the patient was discharged (e.g., home, another facility).
insurance	text	Type of insurance covering the patient.
language	text	Recorded primary language of the patient.
marital_status	text	Patient's marital status as documented.
race	text	Race documented at admission.
sex	text	Sex of the patient.
edatime	text	Datetime the patient was registered in the emergency department (if applicable).
edouttime	text	Datetime the patient left the emergency department.
hospital_expire_flag	bigint	Indicates if patient died during the admission (1) or survived to discharge (0).

Keys and Links

- subject_id → patients.subject_id
- hadm_id is unique within this table (defining each admission).

• d_hcpcs

Column Name	Data Type	Description
code	text	HCPCS code (four characters). Primary key for this table.
category	double precision	Group numeric category for the code.
long_description	text	Longer descriptive text for the code.
short_description	text	Short descriptive text for the code.

Keys and Links

- Joins to hcpcsevents on hcpcs_cd = d_hcpcs.code.

• d_icd_diagnoses

Column Name	Data Type	Description
icd_code	text	ICD diagnosis code. Together with icd_version, this forms a unique definition of a diagnosis.
icd_version	bigint	ICD version (9 or 10).
long_title	text	Full descriptive title for the ICD code (e.g., "Cholera due to vibrio cholerae").

Keys and Links

- Joins to diagnoses_icd on (icd_code, icd_version).

• d_icd_procedures

Column Name	Data Type	Description
icd_code	text	ICD procedure code. Together with icd_version, forms a unique definition of a procedure.
icd_version	bigint	ICD version (9 or 10).
long_title	text	Descriptive name for the procedure code.

Keys and Links

- Joins to procedures_icd on (icd_code, icd_version).

• d_labitems

Column Name	Data Type	Description
itemid	bigint	Primary key. Unique identifier for a laboratory concept.
name	text	Human-readable description of the lab test (e.g., "Sodium").
unit	text	The standard unit for the measurement (e.g., g/L, mmol/L, etc.).
category	text	Higher-level grouping for the lab measurement (e.g., "Chemistry", "Hematology", "ABO").

Keys and Links

- Joins to labevents on itemid.

• diagnoses_icd

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
diag_code	text	The diagnosis code associated with the stay.
description	text	Description of the DRG code.
diag_severity	double precision	Severity of illness rating linked to the DRG.
diag_mortality	double precision	Mortality risk rating linked to the DRG.

Keys and Links

- Joins to d_icd_diagnoses on (icd_code, icd_version).

• diagnoses

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
diag_code	text	The DRG category code.
diag_code	text	DRG code associated with the stay.
description	text	Description of the DRG code.
diag_severity	double precision	Severity of illness rating linked to the DRG.
diag_mortality	double precision	Mortality risk rating linked to the DRG.

• emar

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	double precision	Foreign key referencing admissions.
emar_id	text	Unique identifier for the medication administration.
emar_text	text	Conservative message that orders added to the order entry.
pos_id	text	Foreign key referencing pos. Links to the provider order.
pharmacy_id	double precision	Foreign key referencing pharmacy.
enter_provider_id	text	De-identified ID of the provider entering the eMAR. <i>is deidentified string representing the provider who entered the medication administration information into the database</i>
charttime	text	Time the medication administration was charted.
medication	text	Name of the medication administered.
event_id	text	High-level key about administration status (e.g., "Discontinued", "Not Given").
schedatetime	text	Time the administration was scheduled, if available.
storetime	text	Time the administration was documented in the system.

• emar_detail

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
emar_id	text	Foreign key referencing emar. Unique ID for each order.
emar_seq	text	Consecutive integer which sorts eMAR entries.
patient_id_order_id	double precision	Identifies multiple secondary items for the same eMAR record.
administration_type	text	Type of administration (e.g., "IV Bolus", "Medication Infusion", etc.).
pharmacy_id	double precision	Foreign key referencing pharmacy.
barcode_type	text	Type of barcode scanned (if any).
reason_for_no_barcode	text	Reason no barcode was scanned if barcode scanning wasn't used.
complete_dose_not_g	text	Indicator if the full dose was not given.
dose_dish	text	The expected dose and units for the administration.
dose_given	text	The actual dose administered and units.
dose_given_unit	text	The unit for the product amount.
product_unit	text	Code/description for the scanned product.
prior_infusion_rate	double precision	Infusion rate prior to this administration.
infusion_rate	double precision	The new infusion rate.
infusion_rate_adjustm	double precision	Indicates if the rate has been adjusted.
infusion_rate_adjustm	double precision	Amount of the rate adjustment.
infusion_rate_unit	text	The unit for the infusion rate (e.g., mL/hr).
route	text	Route of administration (e.g., IV, Oral).
infusion_complete	text	Whether the infusion completed as planned.
completion_intent	text	Reason for the infusion completion.
next_to_follow_flag	text	Indicates if a new IV bag was hung.
continued_infusion_in	text	If the infusion continued after patient transfer.
restart_interval	text	Interval between restarting an infusion.
side_effect	text	Documented side effect of administration (e.g., left arm).
non_formulary_flag	text	Documents if a non-formulary medication was visually verified.

• hcpcsevents

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
chartdate	text	Date associated with the coded event.
seq_num	bigint	The HCPCSPT code join within a single admission.
short_description	text	Short textual description for the code.

• labevents

Column Name	Data Type	Description
labevent_id	bigint	Unique identifier for each row in labevents.
subject_id	bigint	Foreign key referencing patients.
hadm_id	double precision	Foreign key referencing admissions.
examtime_id	bigint	Identifier grouping measurements from the same specimen.
itemid	bigint	Lab item tested, join with d_labitems.itemid.
order_provider_id	text	Provider who ordered the lab test (de-identified).
charttime	text	Time the sample was taken or measurement made.
storetime	text	Time the measurement was finalized and stored in the database.
value	text	The lab test result in text form.
valuenum	double precision	Numeric value of the lab test result if applicable.
valuenum	text	Unit of measurement.
ref_range_lower	double precision	Lower bound of the normal range for this lab test, if known.
ref_range_upper	double precision	Upper bound of the normal range for this lab test, if known.
flag	text	Indicates if the result is abnormal or out of normal range.
priority	text	Lab priority (e.g., STAT, Urgent).
comments	text	Additional notes regarding the measurement. Deidentified if sensitive.

• microbiologyevents

Column Name	Data Type	Description
microevent_id	bigint	Unique identifier for each row in microbiologyevents.
subject_id	bigint	Foreign key referencing patients.
hadm_id	double precision	Foreign key referencing admissions.
micro_specimen_id	bigint	Identifier grouping measurements from the same microbiology specimen.
order_provider_id	text	Provider ID who ordered the specimen.
charttime	text	Date of the microbiology observation.
charttime	text	Time of the observation if available (NULL if only the date is known).
spec_itemid	bigint	Item ID describing the type of specimen.
spec_type_desc	text	Description of the specimen (e.g., "Blood", "Urine").
test_seq	bigint	Sequence number differentiating multiple tests on the same specimen.
storetime	text	Date when the result was finalized.
storetime	text	Datetime when the result was finalized.
test_itemid	bigint	ID corresponding to the test performed.
test_name	text	Name of the microbiology test (e.g., "Gram Stain", "Culture").
org_name	double precision	ID of the organism that grew (if any).
org_itemid	text	Name of the organism.
isolate_num	double precision	The isolate number within the specimen (multiple isolates).
quantity	double precision	Approximate quantity of organism grown.
ab_itemid	double precision	ID of the antibiotic tested against the organism.
ab_name	text	Name of the antibiotic tested.
duration_test	text	Free-text describing antibiotic dilution.
duration_comparison	double precision	Comparison operator for the duration (e.g., ">", "<").
duration_value	double precision	Measured duration value.
interpretation	text	Interpretation of antibiotic sensitivity (S = sensitive, R = resistant, I = intermediate, etc.).
comments	text	Additional deidentified text about the culture or sensitivity.

• omr

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
chartdate	text	Date on which the observation was recorded.
seq_num	bigint	Monotonically increasing integer to distinguish multiple similar measurements on the same day.
result_name	text	Description of the observation (e.g., "Blood Pressure", "Weight", "Vital Signs").
result_value	text	The value associated with the observation (e.g., "120/80", "170 cm", "60 kg").

• patients

Column Name	Data Type	Description
subject_id	bigint	Primary key. Unique patient identifier across the dataset.
gender	text	Documented gender.
anchor_age	bigint	Age of the de-identified anchor year (ranged at 90 to 99).
anchor_year	bigint	Billed year of the patient's anchor date.
anchor_year_group	text	Range of years during which the anchor_year occurred (e.g., "2008 - 2010").
dob	text	De-identified date of birth for out-of-hospital mortality linkage (if within one year of hospital discharge).

• pharmacy

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
pharmacy_id	bigint	Primary key within the pharmacy table, used to link to prescriptions or emar.
pos_id	text	Reference to an order in pos.
starttime	text	Start time for the prescribed medication.
stoptime	text	Stop time for the prescribed medication.
medication	text	Name of the medication administered.
pos_type	text	Type of order (e.g., "IV Piggyback", "Time Dosed").
status	text	Whether the prescription is active, discontinued, etc.
entertime	text	Time at which the prescription was entered in the pharmacy system.
verifedtime	text	Time at which the order was verified.
route	text	Route of administration (e.g., IV, Oral).
frequency	text	Frequency of administration (e.g., "QID", "BID").
disp_sched	text	Type(s) of day for scheduled administration (e.g., "08, 20").
infusion_type	text	Code for type of infusion (e.g., "R", "C", "N").
sliding_scale	text	Indicates whether a sliding scale is used (Y/N).
lockout_interval	double precision	Interval for patient-controlled analgesia (time between possible self-administered doses).
basal_rate	double precision	Basal continuous administration rate.
one_hr_max	double precision	Maximum one-hour dose for the medication.
doses_per_24_hrs	double precision	Number of doses expected per 24 hours.
duration	double precision	Numeric duration for this prescription.
duration_interval	text	Unit associated with the duration, e.g., "Doses", "Weeks".
expiration_value	double precision	Numeric length of time until medication expires.
expiration_unit	text	Unit for the medication expiration (days, hours, etc.).
expirationdate	double precision	The expiration date of the medication.
dispensation	text	Source of dispensation.
fill_quantity	double precision	Quantity dispensed.

• pos

Column Name	Data Type	Description
pos_id	text	Primary key within pos. Uniquely identifies the provider order (format: subject_id-DR-SEQ).
pos_seq	bigint	Monotonically increasing sequence for the order.
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
ordertime	text	Time at which the order was placed.
order_type	text	Broad classification of the order (e.g., Lab, Medications, Radiology).
order_subtype	text	More specific subtype of the order (e.g., "Vital Monitor" under "Cardiology").
transcription_type	text	Action performed (e.g., "New", "Change", "Discontinue").
discontinuation_of_pos_id	text	If this order discontinued a previous order, references that pos_id.
discontinuation_of_pos_id	text	If this order was discontinued by another future order, references that future pos_id.
order_provider_id	text	Provider who placed this order (de-identified).
order_status	text	Whether the order is active or has been discontinued.

• pos_detail

Column Name	Data Type	Description
pos_id	text	Foreign key referencing pos. Unique identifier for the order.
pos_seq	bigint	Sequence number for the order.
subject_id	bigint	Foreign key referencing patients.
field_name	text	The label/name of the attribute for this order detail (e.g., "Admit to", "Discharge", etc.).
field_value	text	The value of that attribute (e.g., "Psychiatry", "Clear Liquid Diet").

• prescriptions

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
pharmacy_id	bigint	Foreign key referencing pharmacy.
pos_seq	text	Links to pos.
order_provider_id	double precision	Monotonically increasing sequence for the prescription order.
starttime	text	Provider who placed the prescription order (de-identified).
stoptime	text	Time the prescription was stopped.
drug_type	text	Indicates if the drug is a main component, base, or additive.
formulary_drug_id	text	Name of the medication.
ref_id	double precision	Hospital-specific code for the medication.
formulary_drug_id	text	Generic Sequence Number (e.g., internal medication coding system).
strength	double precision	Strength/concentration description (e.g., "12.5 mg Tablet").
pres_strength	text	Container format for the dose (e.g., "TABLET", "VAL").
dose_val_rx	text	Prescribed dose value.
dose_unit_rx	text	Prescribed dose unit.
form_val_disp	text	Amount of medication contained in a single formulary dose.
form_unit_disp	text	Unit for the formulary dosage.
doses_per_24_hrs	double precision	Number of doses expected in 24 hours.
route	double precision	Route of administration (IV, Oral, etc.).

• procedures_icd

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
seq_num	bigint	The priority of the procedure's billing code.
chartdate	text	Date associated with the billed procedure.
icd_code	text	ICD code representing the procedure.
icd_version	text	ICD version (9 or 10).

Keys and Links

- Joins to d_icd_procedures on (icd_code, icd_version).

• provider

Column Name	Data Type	Description
provider_id	text	Primary key. Unique de-identified provider ID used throughout hosp.

• services

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
transfertime	text	Time the service assignment changed from prev_service to cur_service.
prev_service	text	The previous service (if any).
cur_service	text	The current service (e.g., "SURG", "MED", "PD/CH" etc).

• transfers

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	double precision	Foreign key referencing admissions.
transfer_id	bigint	Unique transfer event identifier (often used to build ICU stays).
eventtype	text	Type of transfer event (e.g., "admit", "transfer", "discharge", "ref").
currentunit	text	The unit or ward to which the patient was transferred (ICU type, regular floor, etc.).
intime	text	Datetime the patient entered the unit.
outtime	text	Datetime the patient left the unit.
los	double precision	Length of stay in the ICU (days).

ICU Module

• caregiver

Column Name	Data Type	Description
caregiver_id	bigint	Primary key in the ICU module. De-identified identifier for ICU caregivers.

• chartevents

Column Name	Data Type	Description
subject_id	bigint	Foreign key referencing patients.
hadm_id	bigint	Foreign key referencing admissions.
stay_id	bigint	Foreign key referencing icustays.
caregiver_id	double precision	De-identified caregiver ID (reference to caregiver).
charttime</		